

ZECHARIAH TAN

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EDUCATION

National University of Singapore

Aug. 2026 – Present

Bachelor of Computing in Computer Science

Temasek Polytechnic

Apr. 2021 – May 2024

Diploma in Information Technology with Merit

3.98 GPA

- Course Valedictorian
- AWS Course Bronze Medal
- DBS Major Project Prize
- SCS Skills Mastery Award

EXPERIENCE

WorldSkills Web Technologies Competitor & Coach

Apr. 2024 – Sep. 2024

Temasek Polytechnic

Singapore

- Earned an *International Bronze Medal* in Web Technologies representing Singapore at WorldSkills Lyon 2024
- Developed training materials and coached competitors for WorldSkills Singapore 2025
- Coached a competitor who earned a *Gold Medal* at WorldSkills Singapore and WorldSkills ASEAN 2025

Software Engineer Intern

Oct. 2023 – Feb. 2024

Centre for Strategic Infocomm Technologies (CSIT)

Singapore

- Built 3 Spring Boot microservices and a Next.js dashboard for indexing and viewing document data
- Integrated search, storage, message queues, and database services across the microservice architecture
- Deployed the microservices to Kubernetes and built observability pipelines for monitoring and logging
- Developed unit and integration tests for backend services as project requirements evolved

PROJECTS

YTMusic API | *TypeScript, npm*

Dec. 2021 – Jul. 2024

- Published a type-safe YouTube Music API wrapper for retrieving songs, artists, lyrics and search results
- Grew the package to *2,500+ weekly npm downloads, 160+ GitHub stars and 40+ forks*
- Built runtime validation and automated tests to handle inconsistent upstream API responses

Finpoint | *Laravel, PHP, React, TypeScript, Sqlite*

Apr. 2026 – Present

- Built a personal finance tracker that consolidates transactions across multiple banks
- Categorized imported transactions into financial records for clearer spending analysis

TheRook | *Rust*

Aug. 2025 – Sep. 2025

- Built a chess engine in Rust using bitboards and precomputed attack data for efficient legal move generation
- Implemented Stockfish-backed perf tests to validate move generation across search trees reaching 15M+ nodes

SounDroid V2 | *Flutter, Dart, Express, TypeScript, Firebase*

Apr. 2022 – Aug. 2022

- Built a full-stack mobile music streaming app with synchronized lyrics, offline downloads, and queue management
- Built user flows for authentication, account management, playlists, and search history

TECHNICAL SKILLS

Languages: TypeScript, Rust, Java, PHP, Dart, SQL

Frameworks: React, Next.js, Flutter, Laravel, Spring Boot, Express, Tailwind CSS

Databases: Elasticsearch, MySQL, PostgreSQL, Sqlite, Firestore

Cloud & Hosting: AWS, Vercel, Firebase, Cloudflare, Fly.io, Heroku

Infrastructure & DevOps: Docker, Kubernetes, NGINX, GitHub Actions, Prometheus, Grafana